

AGSHARAN SAKTHITHARAN

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PROFILE

Final-year Electrical & Electronic Engineering undergraduate at SLIIT with a strong focus on embedded systems, Industrial IoT, and hardware-software integration. Experienced in ESP32 firmware development, MQTT and Modbus RTU communication protocols, PCB design, and full-stack IoT platform development. Seeking R&D and embedded systems roles where research meets hands-on prototyping.

EDUCATION

Bachelor of Science in Electrical & Electronic Engineering Oct 2022 – Sep 2026 (Expected)

Sri Lanka Institute of Information Technology (SLIIT), Sri Lanka

- **CGPA: 3.15**
- Achieved a 4.0 GPA in Year 4 Semester 1, with distinctions in Embedded Systems, Digital Signal Processing, and Robotics
- Final Year Project: Smart Home Energy Management System (SHEMS) — locally-hosted full-stack IoT platform for real-time energy monitoring and smart home automation
- Member, Rotaract Club of SLIIT (2023–2025)

Cambridge Advanced Level — Physical Sciences

2010 – 2022

OKI International School, Wattala, Sri Lanka

- Awarded Best Overall Results

EXPERIENCE

R&D Engineering Intern

Jun 2025 – Sep 2025

SLTMobitel Digital Labs, Colombo, Sri Lanka

- Developed ESP32 firmware for live IoT product development projects involving MQTT, MQTTS, and Modbus RTU communication protocols
- Designed and implemented IR signal learning and replication systems with EEPROM-based storage and a browser-accessible configuration interface using AsyncWebServer
- Contributed to PCB design review, BOM preparation, component selection, and PCB fabrication vendor evaluation
- Assembled and validated four industrial monitoring devices end-to-end — PCB soldering, enclosure fabrication using AutoCAD drilling layouts, sensor calibration, and field testing
- Worked with K-type thermocouples, IR temperature sensors, and current transformers for industrial machine monitoring applications

Electrical Engineering Intern

Jun 2024 – Sep 2024

Kelani Cables PLC, Kelaniya, Sri Lanka

- Assisted in preventive and breakdown maintenance of cable manufacturing machinery including motors, DC drives, and variable frequency drives (VFDs)
- Performed insulation resistance testing using Megger and supported earth resistance testing and QA high-voltage testing of armored cables
- Gained exposure to PLC wiring, relay ladder logic, and industrial panel wiring in a heavy manufacturing environment

SELECTED PROJECTS

Smart Home Energy Management System (SHEMS) | Raspberry Pi 5 · FastAPI · SQLite · Python · MQTT · Docker · Cloudflare Tunnel · PWA

- Built the complete backend infrastructure of a locally-hosted IoT platform including FastAPI server, SQLite database, and Python automation scripts for appliance control logic
- Implemented MQTT subscription pipeline to receive telemetry from distributed ESP32 monitoring nodes and persist data to the database
- Developed a Progressive Web App dashboard for real-time energy visibility and remote appliance control, deployed via Docker Compose with Cloudflare Tunnel

Smart AC Controller | *ESP32 · MQTT/MQTTS · IR Communication · EEPROM · AsyncWebServer · Node-RED*

- Implemented IR signal capture, decoding, and EEPROM-based storage for AC remote control functions using TSOP1738 receiver and IRremoteESP8266 library
- Built a web-based configuration interface on ESP32 with AP/STA dual-mode networking and integrated MQTTS for secure remote control

Industrial Machine Monitoring System | *ESP32 · MAX6675 · K-type Thermocouple · MQTT · PCB Design · AutoCAD*

- Designed and deployed four independent field-ready industrial monitoring devices for tracking machine temperatures and power consumption in a factory environment
- Reviewed and finalized PCB layouts with attention to trace width and component placement for industrial reliability
- Assembled complete devices including PCB soldering, MAX6675 thermocouple module integration, continuity testing, and enclosure fabrication using AutoCAD drilling templates
- Calibrated K-type thermocouples and IR temperature sensors and documented calibration procedures for future reference
- Performed end-to-end system validation with current transformers connected to live power lines and telemetry publishing via MQTT

Egg Crack Detection System | *Raspberry Pi 5 · Python · TensorFlow/Keras · VGG16 · OpenCV*

- Trained a VGG16 CNN model for binary crack classification and integrated IR sensor triggering and relay-based conveyor control on Raspberry Pi 5
- Achieved 72.3% accuracy on a prototype dataset; system demonstrated viable edge inference pipeline for industrial quality control

Semantic-Aware Image Compression with Text-Preserving Block DCT Coding | *MATLAB · DCT · Vector Quantization · LBG Algorithm · OCR*

- Implemented OCR-based semantic masking to classify 8×8 DCT blocks as text or background, applying high-fidelity coefficient preservation for text and LBG-trained vector quantization for background
- Achieved near-lossless text reconstruction (text-region PSNR capped at 120 dB) while maintaining useful compression ratios on background regions

Other Projects: Line Follower Robot with Fuzzy Logic Control (ESP32, TB6612FNG, Sugeno-style inference); Server Room Environmental Monitoring System (ESP32, multi-point temperature and humidity acquisition, LabVIEW visualization).

TECHNICAL SKILLS

Microcontrollers & Firmware: ESP32, STM32, FreeRTOS, Embedded C/C++, Arduino IDE, PlatformIO

Communication Protocols: MQTT/MQTTS, Modbus RTU, RS485, IR Communication, UART, SPI, I2C, WebSocket

IoT & Backend: Raspberry Pi, FastAPI, SQLite, Python, Docker, Cloudflare Tunnel, Mosquitto

Hardware & Electronics: PCB Design (EasyEDA), Circuit Assembly & Soldering, Sensor Interfacing, AutoCAD, Buck Converter Design

Software & Tools: MATLAB, LabVIEW, Quartus II (FPGA/VHDL), NI Multisim, Proteus, Git

Languages: English (Fluent), Tamil (Native), Sinhala (Conversational)

REFERENCES

Available upon request.